

SHENYANG, China

WMO: 543420

Lat: 41.7325N

Lon: 123.5099E

Elev: 49

StdP: 100.74

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-20.4	-28.4	0.3	-17.1	-26.3	0.3	-16.8	8.6	-7.0	7.4	-6.5	1.4	70	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	31.9	23.3	30.7	22.7	29.7	22.2	25.8	29.7	25.0	28.5	24.2	27.5	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.8	28.1	23.9	18.8	27.2	23.1	18.0	26.6	80.2	29.6	76.7	28.7	73.6	27.6	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.4	6.4	-25.3	33.7	3.0	1.5	-27.4	34.7	-29.1	35.6	-30.8	36.4	-33.0	37.5		
			-25.5	26.9	2.9	1.0	-27.6	27.6	-29.3	28.2	-30.9	28.8	-33.1	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-11.3	-6.3	1.6	10.5	17.5	22.2	25.0	23.8	18.1	10.0	0.3	-8.2
	DBStd	13.21	4.72	5.37	4.82	4.49	3.43	2.57	2.00	2.47	3.25	4.50	5.61	5.21
	HDD10.0	2343	659	456	264	47	1	0	0	1	56	294	566	
	HDD18.3	4087	918	689	520	236	56	3	0	1	42	258	541	824
	CDD10.0	1862	0	0	2	63	233	366	464	429	244	57	3	0
	CDD18.3	566	0	0	0	2	31	118	206	172	36	1	0	0
	CDH23.3	4723	0	0	0	26	329	953	1754	1364	288	8	0	0
	CDH26.7	1355	0	0	0	2	69	284	562	400	37	1	0	0
Wind	WSAvg	2.6	2.2	2.6	3.1	3.6	3.3	2.6	2.3	2.1	2.3	2.6	2.7	2.3
Precipitation	PrecAvg	676	7	9	16	40	58	90	171	158	65	44	18	10
	PrecMax	971	28	37	70	105	176	197	364	357	188	142	74	32
	PrecMin	255	0	0	0	0	0	26	35	13	11	1	1	0
	PrecStd	142	6	9	15	24	34	41	85	89	35	29	13	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	9.7	18.0	26.3	30.7	33.2	33.7	32.9	29.5	24.4	17.2	7.1
		MCWB	0.3	3.7	9.8	14.0	18.0	20.4	24.7	24.6	21.2	16.2	11.2	4.3
	2%	DB	1.1	6.6	14.5	23.3	28.5	31.1	32.1	31.2	27.7	22.0	14.0	4.2
		MCWB	-1.7	2.0	7.1	12.9	16.8	20.0	24.5	24.2	19.4	14.9	9.4	1.4
	5%	DB	-0.9	4.6	12.0	21.3	26.9	29.8	30.9	30.2	26.4	19.9	11.7	2.2
		MCWB	-3.3	0.7	5.6	11.6	15.7	19.7	23.9	23.6	18.5	13.8	7.7	-0.2
	10%	DB	-3.0	2.5	9.7	19.0	25.1	28.2	29.7	29.0	25.0	17.9	9.3	0.3
		MCWB	-5.1	-0.8	4.1	10.1	14.8	19.5	23.4	23.0	17.8	12.3	5.9	-2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	4.7	10.5	16.5	20.0	23.2	27.0	26.7	23.3	17.8	12.0	4.9
		MCDB	3.0	8.0	17.2	22.8	28.4	29.0	31.2	30.3	27.2	22.7	15.7	7.0
	2%	WB	-1.5	2.6	8.1	14.0	18.3	22.3	26.0	25.8	21.7	16.0	9.9	1.8
		MCDB	0.6	5.9	13.4	21.5	25.7	27.8	30.4	29.3	25.0	20.1	13.3	3.7
	5%	WB	-3.2	0.9	6.1	12.5	17.1	21.5	25.2	25.1	20.4	14.7	8.2	0.0
		MCDB	-1.1	4.2	11.4	19.6	24.1	26.8	29.0	28.5	24.2	18.8	11.2	2.0
	10%	WB	-5.0	-0.7	4.5	11.1	16.2	20.8	24.5	24.3	19.2	13.4	6.2	-1.9
		MCDB	-3.0	2.4	9.2	17.7	23.0	25.8	28.0	27.5	23.1	17.2	9.1	0.2
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.1	10.3	11.6	11.5	9.7	8.0	8.6	11.6	11.5	9.7	10.2
		MCDBR	13.2	12.5	13.8	15.3	14.6	12.4	9.9	10.2	13.1	13.9	12.9	11.8
		MCWBR	10.9	8.7	8.3	7.5	6.0	4.5	3.9	4.2	5.7	7.5	8.9	9.8
	5% WB	MCDBR	12.8	11.6	12.9	13.0	11.9	9.6	7.8	7.8	10.2	12.0	11.6	11.1
		MCWBR	11.0	8.6	8.3	7.1	5.6	4.1	3.8	4.0	5.4	7.2	8.8	9.6
Clear-Sky Solar Irradiance	taub	0.374	0.440	0.498	0.600	0.651	0.728	0.707	0.619	0.541	0.552	0.465	0.395	
	taud	2.013	1.866	1.802	1.650	1.601	1.527	1.604	1.761	1.865	1.759	1.891	1.993	
	Ebn at Noon	722	725	734	691	670	620	628	670	690	604	605	651	
	Edh at Noon	118	161	193	241	259	280	257	213	179	178	134	112	
All-Sky Solar Radiation	RadAvg	2.36	3.22	4.31	5.10	5.79	5.29	4.84	4.74	4.41	3.28	2.26	1.91	
	RadStd	0.15	0.21	0.28	0.33	0.45	0.52	0.39	0.38	0.26	0.21	0.22	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page